

Translated

From: e-maxx.ru

Burnside's lemma / Pólya enumeration theorem

Burnside's lemma

Burnside's lemma was formulated and proven by **Burnside** in 1897, but historically it was already discovered in 1887 by **Frobenius**, and even earlier in 1845 by **Cauchy**. Therefore it is also sometimes named the **Cauchy-Frobenius lemma**.

Burnside's lemma allows us to count the number of equivalence classes in sets, based on internal symmetry.

Objects and representations

We have to clearly distinguish between the number of objects and the number of representations.

Different representations can correspond to the same objects, but of course any representation corresponds to exactly one object. Consequently the set of all representations is divided into equivalence classes. Our task is to compute the number of objects, or equivalently, the number of equivalence classes. The following example will make the difference between object and representation clearer.

Example: coloring of binary trees

Suppose we have the following problem. We have to count the number of ways to color a rooted binary tree with n vertices with two colors, where at each vertex we do not distinguish between the left and the right children.

Here the set of objects is the set of different colorings of the tree.

We now define the set of representations. A representation of a coloring is a function $f(v)$, which assigns each vertex a color (here we use the colors 0 and 1). The set of representations is the set containing all possible functions of this kind, and its size is obviously equal to 2^n .

At the same time we introduce a partition of this set into equivalence classes.